

Exploratory Data Analysis (EDA) in Python - Detailed Guide

1. Introduction to EDA

Exploratory Data Analysis (EDA) is a critical step in data analysis and machine learning. It involves examining datasets in detail to understand their structure, patterns, relationships, and anomalies before applying any models.

EDA helps in making informed decisions, selecting relevant features, and improving overall model performance.

It uses both statistical summaries and visual techniques to explore the data.

2. Types of Data in EDA

Data can be broadly classified into two categories:

1. Numerical Data: Includes measurable quantities.

- Discrete Data: Countable values (e.g., number of students).
- Continuous Data: Measurable values (e.g., height, weight).

2. Categorical Data: Represents labels or categories.

- Nominal Data: No order (e.g., gender, color).
- Ordinal Data: Has order (e.g., grades A, B, C).

Example: Age (numerical), Department (categorical).

3. Step-by-Step EDA Process

Step 1: Import Libraries (pandas, numpy, matplotlib, seaborn).

Step 2: Load Dataset using `pd.read_csv()`.

Step 3: Inspect Data using `df.head()`, `df.info()`.

Step 4: Handle Missing Values using `fillna()` or `dropna()`.

Step 5: Perform Statistical Analysis using `describe()`.

Step 6: Visualize Data using plots.

Step 7: Detect Outliers and relationships.

4. Measures of Central Tendency

Central tendency describes the center of data.

Mean: Average of values. Formula = $\text{sum}(\text{values})/n$.

Median: Middle value when data is sorted.

Mode: Most frequent value.

Example: Data = [2,4,4,6,8]

Mean = 4.8, Median = 4, Mode = 4.

Python Example:

```
import numpy as np
np.mean([2,4,4,6,8])
```

5. Measures of Dispersion

Dispersion describes how spread out the data is.

Range: Difference between max and min.

Variance: Average squared deviation from mean.

Standard Deviation: Square root of variance.

Example: Data = [1,2,3,4]

Range = 3.

Python: `np.std([1,2,3,4])`

6. Data Distribution and Skewness

Distribution shows how data values are spread.

Normal Distribution is symmetric.

Positive Skew: Tail on right, mean > median.

Negative Skew: Tail on left, mean < median.

Example: Income data is usually positively skewed.

7. Five Number Summary and Box Plot

Five number summary includes:

Minimum, Q1 (25%), Median (50%), Q3 (75%), Maximum.

It helps identify spread and outliers.

Box plot visually represents this summary.

8. Correlation and Covariance

Covariance shows direction of relationship between variables.

Correlation shows strength and direction (-1 to +1).

Example: Study hours vs marks.

Python: `df.corr()`

9. Missing Values Handling

Missing values occur when data is incomplete.

Detection: `df.isnull().sum()`

Handling:

- Fill using mean/median
- Drop rows/columns

Example: `df.fillna(df.mean(), inplace=True)`

10. Outlier Detection

Outliers are extreme values that differ significantly.

Methods:

- IQR method
- Z-score method

Visualization using boxplot.

Example: `sns.boxplot(data=df)`

11. Key EDA Visualizations

Histogram: Distribution of data.

Boxplot: Detect outliers.

Scatter plot: Relationship between variables.

Heatmap: Correlation matrix.

Example:

```
import seaborn as sns
sns.histplot(data)
```

12. Complete Example with Solution

```
import pandas as pd

data = {'Age':[20,21,None,23],'Marks':[85,90,88,92]}
df = pd.DataFrame(data)
df.fillna(df['Age'].mean(), inplace=True)
print(df.describe())
```

This example demonstrates handling missing values and summarizing data.